TankGuard C1 — TRL 7 evidence note

Document: TG-TRL-2026-001 · **Date:** 31 October 2026 · **Prepared for:** TIH IoT Technology Development Program 2026 (TDP 11)
Prepared by: Dr. Vishnu Vasudev (PI), with Nuke Labs

1. Claim

> The C1 cleaning arm has reached **TRL 7** on the drive, structure and manual control axis. On 31 October 2026 the arm was assembled, lowered into a real 500 L water tank, powered from its own supply, and driven in both directions on both joints by the operator through the controller's own web page.

The claim is deliberately narrow. It covers the arm as a machine in its real operating environment. It does not claim the autonomous cleaning cycle, the brush and pump under sludge, or coverage, which the project delivers.

2. The scale used

Definitions as given in the TDP application form.

Level	Definition	Where TankGuard stands
TRL 6	The technology works in near-real or simulated conditions	The arm was previously exercised on the bench, out of the tank
TRL 7	**The technology works in a real place**	**The arm ran in a 500 L tank, in water, under its own power**
TRL 8	The product is finished and ready	Target of this project
TRL 9	People use it every day for real work	After first sales

3. What was demonstrated

Test article. The C1 arm: two rigid 20 cm links, one joint at the shoulder and one at the elbow, a ring brush head at the tip, driven by two 12 V planetary gearmotors through dual-PWM drivers. Controller: ESP8266 NodeMCU running `firmware/c1_cleaner`, powered from its own 12 V supply, with the tank unit not connected.

Environment. A 500 L water storage tank, filled, with the arm mounted at the lid opening and the links immersed. This is the environment the product is sold for, not a bench mock-up.

Procedure.

- Power the controller and join its hotspot.
- Open the controller page at 192.168.4.1.
- In manual mode, jog the shoulder forward and back in bursts, then the elbow forward and back, watching the links in the tank.
- Confirm direction, response and freedom from binding.

Result. Both joints moved in both directions, in the tank, under load. The arm responded to each jog command, reversed on command, and did not bind or stall. Manual control worked end to end: browser, controller, drivers, motors, mechanism, in water.

4. What this proves, and what it does not

Proved.

- The arm's structure and joints survive and operate in a real tank.
- The drivetrain has enough torque to move the links in water.
- The controller, drivers and wiring work in the real installation.
- Direction and speed control respond as commanded, in both directions.

Not yet proved, and not claimed.

- The autonomous arc cycle in the tank (the planner has run in simulation and on the bench).
- Brush and pump behaviour against settled sludge.
- Floor coverage measured in the tank.
- Repeatability across a full three-ring cycle.
- Certification and field reliability.

These are the deliverables of the funded project, and they are what separate TRL 7 from TRL 8.

5. Why this is TRL 7 and not TRL 6

TRL 6 is a demonstration in a near-real or simulated environment: a bench rig, a dry test stand, a model tank. That is what the arm had before this test.

TRL 7 is a demonstration in the real place. The arm was not on a bench; it was in a 500 L tank, in water, doing the job it is designed to do, with the hardware that will be installed. The distinction is the environment, and the environment is exactly what changed.

6. Path from TRL 7 to TRL 8

Milestone	What closes the gap
M1, T0 + 3 months	Line installed, design frozen, certification scope agreed
M2, T0 + 6 months	Twenty-five pilot sets built in-house; IP68 and EMC passed; unit cost confirmed
M3, T0 + 9 months	Twelve installations in three cities; sixty days of operation
M4, T0 + 12 months	Certification filed, three paying pilots, channel agreement, TRL 8 evidence pack

7. Evidence held

- Photographs and video of the arm running in the 500 L tank, uploaded separately with the application.
- Serial log from the controller covering the manual jog commands.
- This note, signed by the PI and the industry partner.

Dr. Vishnu Vasudev
Principal Investigator
Sree Buddha College of Engineering

Nuke Labs
Industry partner